can construct a path from θ_1^A to θ_1^B such that the loss along the path is bounded by $\max\{L(f_{\theta_1^A}), L(f_{\theta_1^B})\}$. The main result then follows from the fact that $L(f_{\theta_1^A}) \le L(f_{\theta^A}) + \epsilon$ and $L(f_{\theta_1^B}) \le L(f_{\theta^B}) + \epsilon$ due to θ^A and θ^B both being ϵ-dropout stable. □

B Proofs for connectivity via noise stability

In this section, we give detailed proofs showing that noise stability implies connectivity. In the following lemma, we first show that the network output is stable if we randomly dropout columns in a single layer using Algorithm 1.

Lemma 5. *For any layer $2 \le i \le d$, let $G = \{(U^{(l)}, x^{(l)})\}_{l=1}^m$ be a set of matrix/vector pairs of size m where $U \in \mathbb{R}^{h_d \times h_i}$ and $x \in \mathbb{R}^{h_{i-1}}$ satisfying $\|x\|_\infty = O\left(\frac{\|x\|}{\sqrt{h_{i-1}}}\right)$. Given A_i, let $\hat{A}_i \in \mathbb{R}^{h_i \times h_{i-1}}$ be the output of Algorithm 1 with dropout probability $0 < p \le \frac{3}{4}$. Assume $\|[A_i]_j\| = O(\sqrt{p})\|A_i\|_F$ for $1 \le j \le h_{i-1}$. Given any $0 < \delta < 1$, let $\epsilon' = O\left(\sqrt{\frac{\log(mh_d/\delta)p}{h_{\min}}}\right)$, with probability at least $1-\delta$, we have for any $(U, x) \in G$ that $\|U(\hat{A}_i - A_i)x\| \le \epsilon'\|A_i\|_F\|U\|\|x\|$. Further assuming $h_{\min} = \Omega\left(\frac{\log(1/\delta)}{p}\right)$, we know with probability at least $1-\delta$, no less than $\frac{2}{3}p$ fraction of columns in $\hat{A}_i$ are zero vectors.*

Intuitively, this lemma upper-bounds the change in the network output after dropping out a single layer. In the lemma, we should think of x as the input to the current layer, A_i as the layer matrix and U as the Jacobian of the network output with respect to the layer output. If the activation pattern does not change after the dropping out, $U\hat{A}_i x$ is exactly the output of the dropped out network and $\|U(\hat{A}_i - A_i)x\|$ is the change in the network output.

Proof of Lemma 5. Fixing $2 \le i \le d$ and one pair $(U, x) \in G$, we show with probability at least $1 - \frac{\delta}{m}$, $\|U(\hat{A}_i - A_i)x\| \le \epsilon'\|A_i\|_F\|U\|\|x\|$. Let U_k be the k-th column of U. Then by definition of $\hat{A}_i$ in the algorithm, we know

$$\begin{aligned} U(\hat{A}_i - A_i)x &= \sum_{k,j} U_k [A_i]_{kj} x_j (\delta_j - 1) \\ &= \sum_j \left(\sum_k U_k [A_i]_{kj}\right) x_j (\delta_j - 1), \end{aligned}$$

where δ_j is an i.i.d. Bernoulli random variable which takes the value 0 with probability p and takes the value $\frac{1}{1-p}$ with probability $(1-p)$.

Let $[A_i]_j$ be the j-th column of A_i. Because $p \le \frac{3}{4}$, $\frac{1}{1-p} = O(1)$ (any p bounded away from 1 will work). Hence the norm for each individual term can be bounded as follows.

$$\begin{aligned} \left\|\left(\sum_k U_k [A_i]_{kj}\right) x_j (\delta_j - 1)\right\| &\overset{(*)}{\le} O\left(\frac{\|x\|}{\sqrt{h_{i-1}}}\right) \|U[A_i]_j\| \\ &\le O\left(\frac{\|x\|}{\sqrt{h_{\min}}}\right) \|U\|\|[A_i]_j\| \\ &\overset{(\dagger)}{\le} O\left(\frac{\sqrt{p}\|U\|\|A_i\|_F\|x\|}{\sqrt{h_{\min}}}\right), \end{aligned}$$

where (*) uses the assumption that $\|x\|_\infty = O\left(\frac{\|x\|}{\sqrt{h_{i-1}}}\right)$ and (†) holds because we assume $\|[A_i]_j\| = O(\sqrt{p})\|A_i\|_F$ for $1 \le j \le h_{i-1}$.

For the total variance, we have

$$\begin{aligned}
\sigma^2 :&= \sum_j \mathbb{E}\left[\left\|\left(\sum_k U_k[A_i]_{kj}\right) x_j(\delta_j - 1)\right\|^2\right] \\
&\le \sum_j \|U[A_i]_j\|^2 |x_j|^2 \left((0-1)^2 \times p + \left(\frac{1}{1-p} - 1\right)^2 \times (1-p)\right) \\
&\overset{(*)}{=} \sum_j \|U[A_i]_j\|^2 \cdot O\left(\frac{\|x\|^2}{h_{i-1}}\right) \cdot p\left(1 + \frac{p}{1-p}\right) \\
&\le \|UA_i\|_F^2 \cdot O\left(\frac{\|x\|^2}{h_{\min}}\right) \cdot p \\
&\le O\left(\frac{p\|U\|^2\|A_i\|_F^2\|x\|^2}{h_{\min}}\right),
\end{aligned}$$

where inequality $(*)$ uses the assumption that $\|x\|_\infty = O\left(\frac{\|x\|}{\sqrt{h_{i-1}}}\right)$. Then, by the vector Bernstein inequality (Lemma 8), we know given $0 < \delta < 1$, there exists $\epsilon' = O\left(\sqrt{\frac{p\log(mh_d/\delta)}{h_{\min}}}\right)$, with probability at least $1 - \frac{\delta}{m}$, we have

$$\|U(\hat{A}_i - A_i)x\| \le \epsilon'\|A_i\|_F\|U\|\|x\|.$$

Taking the union bound over all (U, x) pairs in G, we know that with probability at least $1 - \delta$, for any $(U, x) \in G$, $\|U(\hat{A}_i - A_i)x\| \le \epsilon'\|A_i\|_F\|U\|\|x\|$.

Suppose $h_{\min} = \Omega\left(\frac{\log(1/\delta)}{p}\right)$; then by the Chernoff bound, we know with probability at least $1 - \delta$, the dropped out fraction is at least $\frac{2}{3}p$. Taking another union bound concludes our proof. □

Now we are ready to prove Lemma 3. The idea is similar to (Arora et al., 2018), but we give the proof here for completeness.

Lemma 3. *Let θ be an ϵ-noise stable network, and let θ_1 be the network with weight matrices from layer 2 to layer d dropped out by Algorithm 1 with dropout probability $\widetilde{\Omega}(1/h_{min}) < p \le \frac{3}{4}$. For any $2 \le i \le d$, assume $\|[A_i]_j\| = O(\sqrt{p})\|A_i\|_F$ for $1 \le j \le h_{i-1}$. For any $0 \le t \le 1$, define the network on the segment from θ to θ_1 as $\theta_t := \theta + t(\theta_1 - \theta)$. Then, with probability at least $1/4$ over the weights generated by Algorithm 1, $L(f_{\theta_t}) \le L(f_\theta) + \widetilde{O}(\sqrt{p}\epsilon)$, for any $0 \le t \le 1$.*

Proof of Lemma 3. We first bound the difference between the dropped out network θ_1 and the original network θ.

Bounding $\|f_\theta(x) - f_{\theta_1}(x)\|$: We first show that with probability at least $1/2 - \delta$, $\|f_\theta(x) - f_{\theta_1}(x)\| = \|x^d - \hat{x}_d^d\| \le \epsilon'\|f_\theta(x)\|$, where ϵ' will be specified later. For any layer $i \ge 1$ and letting $\hat{x}_i^j$ be the vector before activation at layer j if the weights $A_2, \ldots, A_i$ are replaced by $\hat{A}_2, \ldots, \hat{A}_i$.

According to Lemma 5, for any layer $2 \le i \le d$, given $0 < \delta < 1$, let $\epsilon' = O\left(\sqrt{\frac{pc^2d^2\log(mdh_d/\delta)}{h_{\min}\min_{2\le i\le d}(\mu_i^2\mu_{i\to}^2)}}\right)$, with probability at least $1 - \delta/d$ over $\hat{A}_i$, we have

$$\|U(\hat{A}_i - A_i)x\| \le \frac{\epsilon'\mu_i\mu_{i\to}}{6cd}\|A\|_F\|U\|\|x\| \tag{1}$$

for any $(U, x) \in \{(J_{x^i}^{i,j}, \phi(\hat{x}_{i-1}^{i-1}))|x \in S, i \le j \le d\}$. By taking a union bound over i, we know inequality (1) holds with probability at least $1 - \delta$ for every i. Recall that the interlayer smoothness holds with probability

at least 1/2. Taking another union, we know with probability at least $1/2 - \delta$, interlayer smoothness holds and inequality (1) holds for every $2 \le i \le d$. Next, conditioning on the success of these two events, we will inductively prove for any $1 \le i \le d$, for any $i \le j \le d$,

$$\|\hat{x}_i^j - x^j\| \le (i/d)\epsilon' \|x^j\|.$$

For the base case $i = 1$, since we are not dropping out any weight matrix, the inequality is trivial. For any $1 \le i-1 \le d-1$, suppose $\|\hat{x}_{i-1}^j - x^j\| \le \frac{i-1}{d}\epsilon'\|x^j\|$ for any $i-1 \le j \le d$; we prove the induction hypothesis holds for layer i.

For any $i \le j \le d$ we have

$$\|\hat{x}_i^j - x^j\| = \|(\hat{x}_i^j - \hat{x}_{i-1}^j) + (\hat{x}_{i-1}^j - x^j)\| \le \|\hat{x}_i^j - \hat{x}_{i-1}^j\| + \|\hat{x}_{i-1}^j - x^j\|.$$

By the induction hypothesis, we know the second term can be bounded by $(i-1)\epsilon'\|x^j\|/d$. Therefore, in order to complete the induction step, it suffices to show that the first term is bounded by $\epsilon'\|x^j\|/d$. For simplicity, we also denote $\hat{x}_{i-1}^{i-1}$ as $\hat{x}^{i-1}$. Let $\Delta_i = \hat{A}_i - A_i$. We can decompose the error into two terms:

$$\begin{aligned}
\|\hat{x}_i^j - \hat{x}_{i-1}^j\| &= \|M^{i,j}(\hat{A}_i\phi(\hat{x}^{i-1})) - M^{i,j}(A_i\phi(\hat{x}^{i-1}))\| \\
&= \|M^{i,j}(\hat{A}_i\phi(\hat{x}^{i-1})) - M^{i,j}(A_i\phi(\hat{x}^{i-1})) + J_{x^i}^{i,j}(\Delta_i\phi(\hat{x}^{i-1})) - J_{x^i}^{i,j}(\Delta_i\phi(\hat{x}^{i-1}))\| \\
&\le \|J_{x^i}^{i,j}(\Delta_i\phi(\hat{x}^{i-1}))\| + \|M^{i,j}(\hat{A}_i\phi(\hat{x}^{i-1})) - M^{i,j}(A_i\phi(\hat{x}^{i-1})) - J_{x^i}^{i,j}(\Delta_i\phi(\hat{x}^{i-1}))\| \qquad (2)
\end{aligned}$$

The first term of (2) can be bounded as follows:

$$\begin{aligned}
&\|J_{x^i}^{i,j}\Delta_i\phi(\hat{x}^{i-1})\| \\
&\le (\epsilon'\mu_i\mu_{i\to}/6cd)\|J_{x^i}^{i,j}\|\|A_i\|_F\|\phi(\hat{x}^{i-1})\| && \text{Lemma 5} \\
&\le (\epsilon'\mu_i\mu_{i\to}/6cd)\|J_{x^i}^{i,j}\|\|A_i\|_F\|\hat{x}^{i-1}\| && \phi \text{ (ReLU) is 1-Lipschitz} \\
&\le (\epsilon'\mu_i\mu_{i\to}/3cd)\|J_{x^i}^{i,j}\|\|A_i\|_F\|x^{i-1}\| && \text{Induction hypothesis,} \\
& && \left\|\hat{x}^{i-1} - x^{i-1}\right\| \le \frac{(i-1)\epsilon'\left\|x^{i-1}\right\|}{d} < \left\|x^{i-1}\right\| \\
&\le (\epsilon'\mu_i\mu_{i\to}/3d)\|J_{x^i}^{i,j}\|\|A_i\|_F\|\phi(x^{i-1})\| && \text{Activation Contraction} \\
&\le (\epsilon'\mu_{i\to}/3d)\|J_{x^i}^{i,j}\|\|A_i\phi(x^{i-1})\| && \text{Layer Cushion} \\
&= (\epsilon'\mu_{i\to}/3d)\|J_{x^i}^{i,j}\|\|x^i\| && x^i = A_i\phi(x^{i-1}) \\
&\le (\epsilon'/3d)\|x^j\| && \text{Interlayer Cushion} \qquad (3)
\end{aligned}$$

The second term of (2) can be bounded as:

$$\begin{aligned}
&\|M^{i,j}(\hat{A}_i\phi(\hat{x}^{i-1})) - M^{i,j}(A_i\phi(\hat{x}^{i-1})) - J_{x^i}^{i,j}(\Delta_i\phi(\hat{x}^{i-1}))\| \\
&= \|(M^{i,j} - J_{x^i}^{i,j})(\hat{A}_i\phi(\hat{x}^{i-1})) - (M^{i,j} - J_{x^i}^{i,j})(A_i\phi(\hat{x}^{i-1}))\| \\
&\le \|(M^{i,j} - J_{x^i}^{i,j})(\hat{A}_i\phi(\hat{x}^{i-1}))\| + \|(M^{i,j} - J_{x^i}^{i,j})(A_i\phi(\hat{x}^{i-1}))\|. \qquad (4)
\end{aligned}$$

Both terms of (4) can be bounded using the interlayer smoothness condition. For the second term of (4), notice that $A_i\phi(\hat{x}^{i-1}) = \hat{x}_{i-1}^i$. Thus by the induction hypothesis, we know

$$\|A_i\phi(\hat{x}^{i-1}) - x^i\| = \|\hat{x}_{i-1}^i - x^i\| \le (i-1)\epsilon'\|x^i\|/d \le \epsilon'\|x^i\|. \qquad (5)$$

Now, by interlayer smoothness,

$$\begin{aligned}
\|(M^{i,j} - J_{x^i}^{i,j})(A_i\phi(\hat{x}^{i-1}))\| &= \|(M^{i,j} - J_{x^i}^{i,j})(x^i + (A_i\phi(\hat{x}^{i-1}) - x^i))\| \\
&\le \frac{\|A_i\phi(\hat{x}^{i-1}) - x^i\|\|x^j\|}{\rho\|x^i\|} \\
&\overset{(*)}{\le} \frac{\epsilon'\|x^i\|\|x^j\|}{3d\|x^i\|} = \frac{\epsilon'\|x^j\|}{3d} \qquad (6)
\end{aligned}$$